

Secure Multi-Branch Banking Network Infrastructure

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LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course
**CSE322: Computer Networks Lab in the Department of Computer
Science and Engineering**



DAFFODIL INTERNATIONAL UNIVERSITY

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DECLARATION

We hereby declare that this lab project has been done by us under the supervision of Mr. **Tanvirul Islam**, Lecturer, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:

Table 1: Course Outcome Statements

CO's	Statements
CO1	Describe and apply the fundamentals of LAN architecture along with basic topology setup using simulation tools.
CO2	Configure and Analyze network devices and services using Cisco Packet Tracer in various networking topologies.
CO3	Design, Implement, and Troubleshoot dynamic and static routing protocols NAT, and ACLs in simulated environments to build efficient and secure networks.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP	CEA
CO1	PO5	C3, P1, P2, A2	K6	EP1	---
CO2	PO2	C3, P2	K1-K4	EP2	---
CO3	PO3	A2, P2	K5	EP3	—

The mapping justification of this table is provided in section 4.3.1, 4.3.2 and 4.3.3.

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Chapter 1

Introduction

This chapter gives a brief overview of the Secure Multi-Branch Banking Network Infrastructure project. The project connects a Head Office with three branches using Cisco Packet Tracer. It uses VLAN, inter-VLAN routing, centralized DHCP, RIP v2, DNS, and HTTP services.

1.1 Introduction

A banking organization needs a reliable network to connect its different branches and departments. In this project, a Head Office and three branches are connected through WAN links. VLANs are used to separate different departments, and Router-on-a-Stick is used for communication between VLANs.

1.2 Motivation

The main purpose of this project is to build a simple and manageable banking network. A centralized DHCP server is used at the Head Office so that all branches can receive IP addresses automatically. RIP v2 is used to maintain communication between different branches.

1.3 Objectives

The main objectives are:

- To connect the Head Office with three branches.
- To separate departments using VLANs.
- To implement inter-VLAN routing.
- To provide centralized DHCP service.
- To configure RIP v2 for dynamic routing.
- To provide DNS and HTTP services.
- To test communication between all branches.

1.4 Feasibility Study

The project is feasible using Cisco Packet Tracer. The required routers, switches, PCs, and server can be configured and tested in the simulation environment. The central server provides DHCP, DNS, and HTTP services for the network.

1.5 Gap Analysis

Managing separate network services at different branches can make the network difficult to maintain. This project solves this problem by using centralized DHCP, dynamic routing, and common DNS and HTTP services. This makes the network easier to manage and connect.

1.6 Project Outcome

The expected outcome is a working multi-branch banking network in Cisco Packet Tracer. The system will support VLANs, inter-VLAN routing, centralized DHCP, RIP v2, DNS, HTTP, and communication between the Head Office and all branches.

Chapter 2

Network Architecture

This chapter describes the network design of the proposed banking system. It explains the main network components, topology, IP addressing, and the simulation tools used to build and test the network.

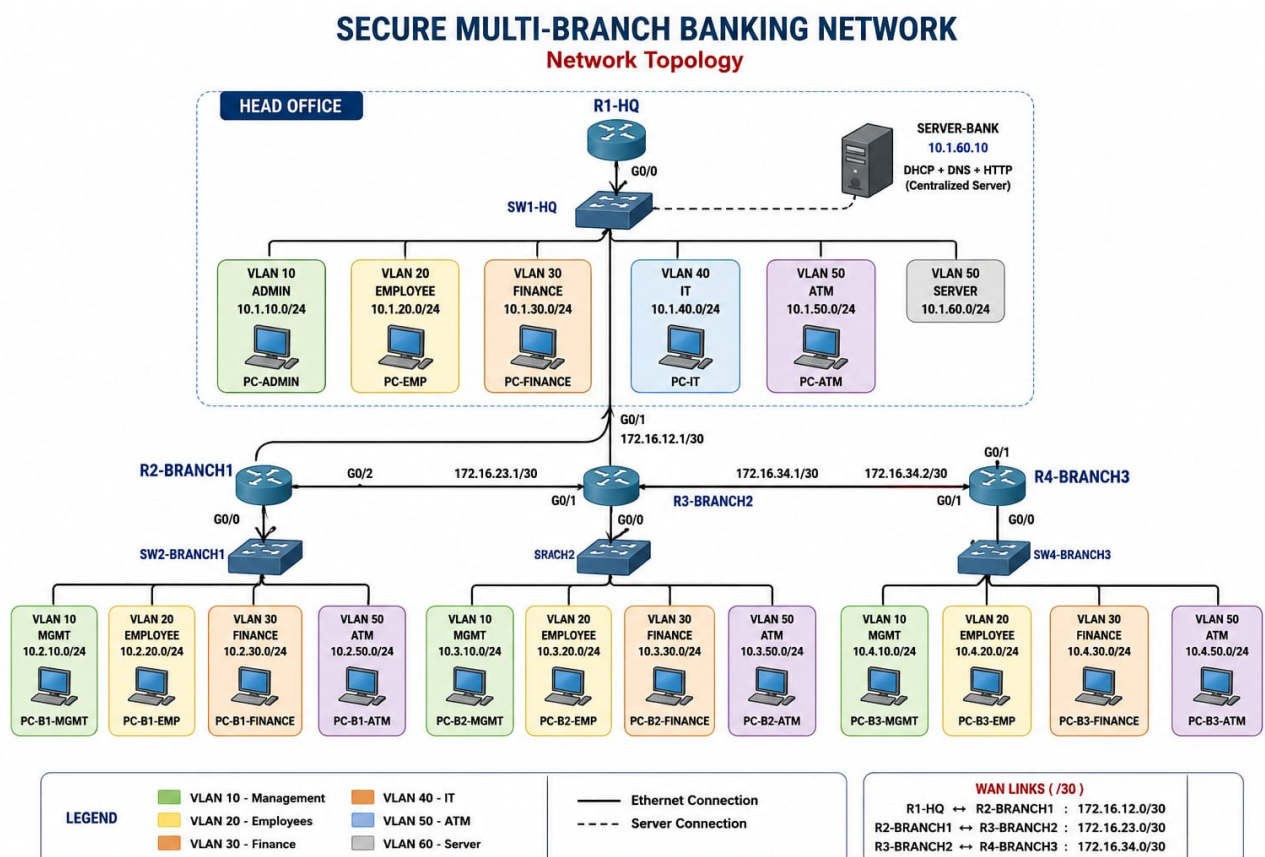
2.1 Requirement Analysis & Design Specification

2.1.1 Requirements

The main requirements of the project are:

- 4 Cisco 2911 routers
- 4 Cisco 2960 switches
- PCs for Head Office and branches
- 1 centralized banking server
- VLANs for different departments
- WAN links between branches
- Centralized DHCP, DNS, and HTTP services
- RIP version 2 for dynamic routing

2.1.2 Network Topology Diagram



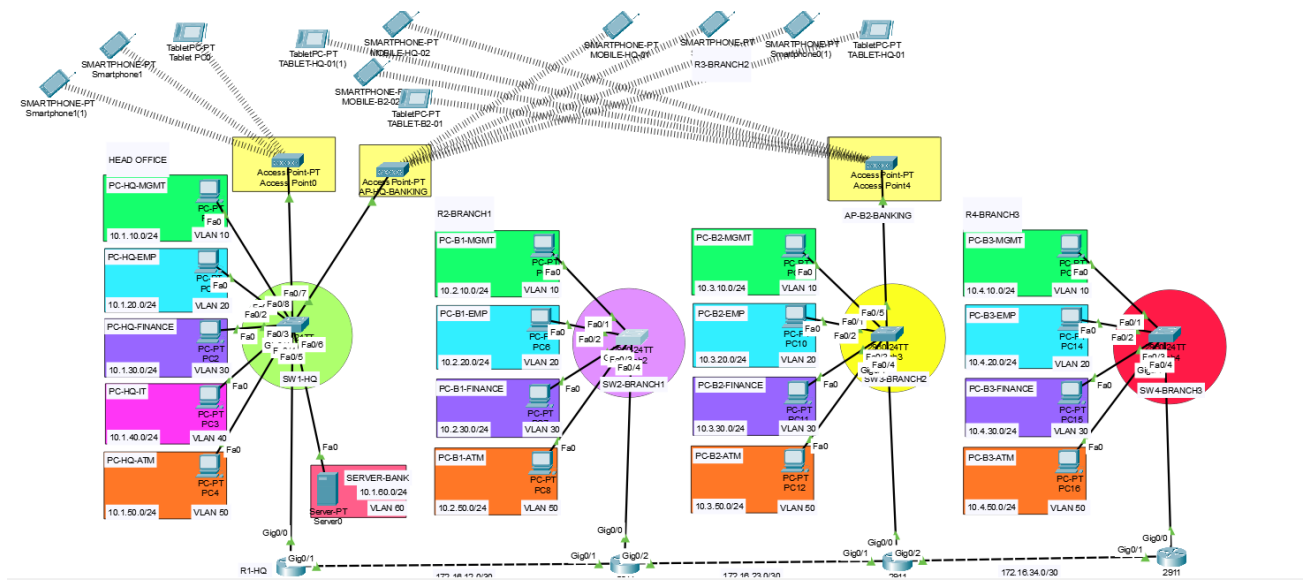


Figure: Network diagram

2.1.3 IP Addressing & Subnetting Table

#	Pool Name	Network	Default Gateway	Start IP	Subnet Mask	Max Users
HQ						
1	HQ-MANAGEMENT	10.1.10.0/24	**10.1.10.1**	**10.1.10.11**	255.255.255.0	50
2	HQ-EMPLOYEES	10.1.20.0/24	**10.1.20.1**	**10.1.20.11**	255.255.255.0	50
3	HQ-FINANCE	10.1.30.0/24	**10.1.30.1**	**10.1.30.11**	255.255.255.0	50
4	HQ-IT	10.1.40.0/24	**10.1.40.1**	**10.1.40.11**	255.255.255.0	50
5	HQ-ATM	10.1.50.0/24	**10.1.50.1**	**10.1.50.11**	255.255.255.0	50
Branch 1						
6	B1-MANAGEMENT	10.2.10.0/24	**10.2.10.1**	**10.2.10.11**	255.255.255.0	50
7	B1-EMPLOYEES	10.2.20.0/24	**10.2.20.1**	**10.2.20.11**	255.255.255.0	50
8	B1-FINANCE	10.2.30.0/24	**10.2.30.1**	**10.2.30.11**	255.255.255.0	50
9	B1-ATM	10.2.50.0/24	**10.2.50.1**	**10.2.50.11**	255.255.255.0	50
Branch 2						
10	B2-MANAGEMENT	10.3.10.0/24	**10.3.10.1**	**10.3.10.11**	255.255.255.0	50
11	B2-EMPLOYEES	10.3.20.0/24	**10.3.20.1**	**10.3.20.11**	255.255.255.0	50
12	B2-FINANCE	10.3.30.0/24	**10.3.30.1**	**10.3.30.11**	255.255.255.0	50
13	B2-ATM	10.3.50.0/24	**10.3.50.1**	**10.3.50.11**	255.255.255.0	50
Branch 3						
14	B3-MANAGEMENT	10.4.10.0/24	**10.4.10.1**	**10.4.10.11**	255.255.255.0	50
15	B3-EMPLOYEES	10.4.20.0/24	**10.4.20.1**	**10.4.20.11**	255.255.255.0	50
16	B3-FINANCE	10.4.30.0/24	**10.4.30.1**	**10.4.30.11**	255.255.255.0	50
17	B3-ATM	10.4.50.0/24	**10.4.50.1**	**10.4.50.11**	255.255.255.0	50

2.2 Use of Modern Simulation Tools

Cisco Packet Tracer was used to design and simulate the complete banking network. It was used to configure routers, switches, VLANs, WAN connections, RIP v2, DHCP relay, DNS, and HTTP services.

Different testing methods were also used to verify the network. Ping was used to check connectivity between routers, branches, and the banking server. Commands such as "show ip route rip", "show vlan brief", and "show interfaces trunk" were used to check routing, VLANs, and trunk connections.

Chapter 3

Implementation and Results

This chapter describes how the banking network was implemented and tested using Cisco Packet Tracer. It also presents the main outputs and the results obtained from the testing process.

3.1 Implementation

The network was implemented by configuring routers, switches, PCs, and a central server. VLANs were created for different departments, and Router-on-a-Stick was used for inter-VLAN communication. RIP v2 was configured for dynamic routing between the Head Office and branches.

The central server was configured to provide DHCP, DNS, and HTTP services. DHCP relay was also configured so that branch PCs could receive IP addresses from the central DHCP server.

3.2 Output

show ip interface brief

```
Router>
Router>
Router>en
Router#
Router#
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	10.2.10.1	YES	manual	up	up
GigabitEthernet0/0.20	10.2.20.1	YES	manual	up	up
GigabitEthernet0/0.30	10.2.30.1	YES	manual	up	up
GigabitEthernet0/0.50	10.2.50.1	YES	manual	up	up
GigabitEthernet0/1	172.16.12.2	YES	manual	up	up
GigabitEthernet0/2	172.16.23.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#|
```

```
Router>
Router>
Router>en
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	10.1.10.1	YES	manual	up	up
GigabitEthernet0/0.20	10.1.20.1	YES	manual	up	up
GigabitEthernet0/0.30	10.1.30.1	YES	manual	up	up
GigabitEthernet0/0.40	10.1.40.1	YES	manual	up	up
GigabitEthernet0/0.50	10.1.50.1	YES	manual	up	up
GigabitEthernet0/0.60	10.1.60.1	YES	manual	up	up
GigabitEthernet0/1	172.16.12.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#|
```

show ip protocols

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 18 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv  Triggered RIP  Key-chain
GigabitEthernet0/1   22
GigabitEthernet0/0.10 22
GigabitEthernet0/0.20 22
GigabitEthernet0/0.30 22
GigabitEthernet0/0.50 22
GigabitEthernet0/2    22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  172.16.0.0
Passive Interface(s):
Routing Information Sources:
  Gateway            Distance      Last Update
--More-- |
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 1 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv  Triggered RIP  Key-chain
GigabitEthernet0/0.10 22
GigabitEthernet0/0.20 22
GigabitEthernet0/0.30 22
GigabitEthernet0/0.40 22
GigabitEthernet0/0.50 22
GigabitEthernet0/1    22
GigabitEthernet0/0.60 22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  172.16.0.0
Passive Interface(s):
Routing Information Sources:
  --More-- |
```

show vlan brief

```
SW1-HQ#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
10	MANAGEMENT	active	Fa0/1, Fa0/7
20	EMPLOYEES	active	Fa0/2, Fa0/8
30	FINANCE	active	Fa0/3
40	IT	active	Fa0/4
50	ATM	active	Fa0/5
60	SERVERS	active	Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1-HQ#
```

```
SW2-BRANCH1#
```

```
SW2-BRANCH1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
10	MANAGEMENT	active	Fa0/1
20	EMPLOYEES	active	Fa0/2
30	FINANCE	active	Fa0/3
50	ATM	active	Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW2-BRANCH1#
```

show interfaces trunk

```
SW1-HQ#
```

```
SW1-HQ#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gig0/1	1-1005

Port	Vlans allowed and active in management domain
Gig0/1	1,10,20,30,40,50,60

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	1,10,20,30,40,50,60

```
SW1-HQ#
```

```

SW2-BRANCH1#
SW2-BRANCH1#show interfaces trunk
Port          Mode          Encapsulation  Status        Native vlan
Gig0/1        on            802.1q         trunking      1

Port          Vlans allowed on trunk
Gig0/1        1-1005

Port          Vlans allowed and active in management domain
Gig0/1        1,10,20,30,50

Port          Vlans in spanning tree forwarding state and not pruned
Gig0/1        1,10,20,30,50

SW2-BRANCH1#

```

3.3 Discussion

The implementation shows that the proposed network can successfully connect the Head Office and three branches. VLANs provide proper department separation, while RIP v2 allows routers to communicate dynamically.

Overall, the project demonstrates a simple and manageable banking network using Cisco Packet Tracer.

Chapter 4

Engineering Standards and Mapping

This chapter describes the engineering practices followed in the project. It also explains the impact of the banking network on society, environment, and sustainability.

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

The proposed banking network helps provide faster and more reliable communication between the Head Office and branches. It can improve the availability of banking services and make daily network-based activities easier.

4.1.2 Impact on Society & Environment

A well-connected banking network can support better communication between branches and reduce unnecessary physical communication. As the project is simulated using Cisco Packet Tracer, it does not require physical networking equipment during development, which reduces resource usage

4.1.3 Ethical Aspects

The project focuses on responsible use of network resources. User information and banking services should be handled securely and only authorized users should be allowed to access sensitive resources.

4.1.4 Sustainability Plan

The network design can be expanded in the future by adding more branches, users, and services. Proper network management and security practices can help maintain the system for a longer period.

4.2 Project Management and Team Work

The project was completed through proper planning and teamwork. The tasks were divided among team members according to their responsibilities. Team members worked together on network design, device configuration, testing, troubleshooting, and report preparation.

Regular communication helped the team identify problems and complete the project within the planned time. Teamwork also helped in sharing technical knowledge and improving the overall quality of the project.

4.3 Complex Engineering Problem

The project involves several networking challenges, including VLAN segmentation, inter-VLAN communication, WAN connectivity, dynamic routing, and centralized services. These components need to work together properly to achieve successful communication between the branches.

4.3.1 Mapping of Program Outcome

The project applies networking knowledge and practical engineering skills to design, configure, and test a multi-branch banking network. It also develops problem-solving, system design, testing, and technical documentation skills.

Table 4.1: Justification of Program Outcomes

POs	Justification of Mapping (Project Perspective)
PO5	The project uses Cisco Packet Tracer to design, configure, test, and analyze the multi-branch banking network in a simulated environment.
PO2	The project requires identifying and solving networking problems related to VLANs, IP addressing, routing, DHCP, and connectivity between branches.
PO3	The project involves designing and implementing a structured network using VLANs, show ip protocols, show vlan brief, RIP v2, and centralized network services to provide reliable communication.

4.3.2 Complex Problem Solving

Table 4.2: Mapping with complex problem solving.

EP1 Dept of Knowledge	EP2 Range of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stakeholder Involvement	EP7 Inter- dependence
✓	✓	✓				

EP1: This project required the design and implementation of a fully functional network topology from scratch. The project requires networking knowledge including VLAN, IP addressing, routing, DHCP, DNS.

EP2: Configuring and analyzing network services across varied topologies requires students to balance a range of conflicting requirements, such as performance versus resource constraints. Using Cisco Packet Tracer allows them to systematically test and optimize device configurations to resolve these competing operational demands.

EP3: Network connectivity, routing tables, VLAN configuration, and DHCP operation are analyzed during testing.

4.3.3 Complex Engineering Activities

Table 4.2: Mapping with complex engineering activities.

EA1	EA2	EA3	EA4	EA5
✓	✓	✓		

EA1: The project involves designing a multi-branch banking network with routers, switches, VLANs, and WAN connections.

EA2: The project requires applying networking concepts such as IP addressing, inter-VLAN routing, RIP v2, DHCP, DNS server.

EA3: The network was configured and tested using Cisco Packet Tracer to verify connectivity and proper data communication.

Chapter 5

Conclusion

This chapter presents the overall conclusion of the Secure Multi-Branch Banking Network Infrastructure project. It summarizes the work, limitations, and possible improvements for the future.

5.1 Summary

The project successfully designed and simulated a multi-branch banking network using Cisco Packet Tracer. The network connects the Head Office with three branches using VLANs, inter-VLAN routing, RIP v2, centralized DHCP, DNS, and HTTP services.

The testing shows that the branches can communicate with each other and access the centralized banking services.

5.2 Limitation

The project was developed and tested in a simulated environment using Cisco Packet Tracer. It does not represent a complete real-world banking network. Advanced security features such as firewall protection, VPN, and advanced access control were not implemented.

5.3 Future Work

In the future, the network can be improved by adding stronger security features, firewall protection, VPN connectivity, and access control. More branches and users can also be added to make the system more scalable and closer to a real banking network.

References

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